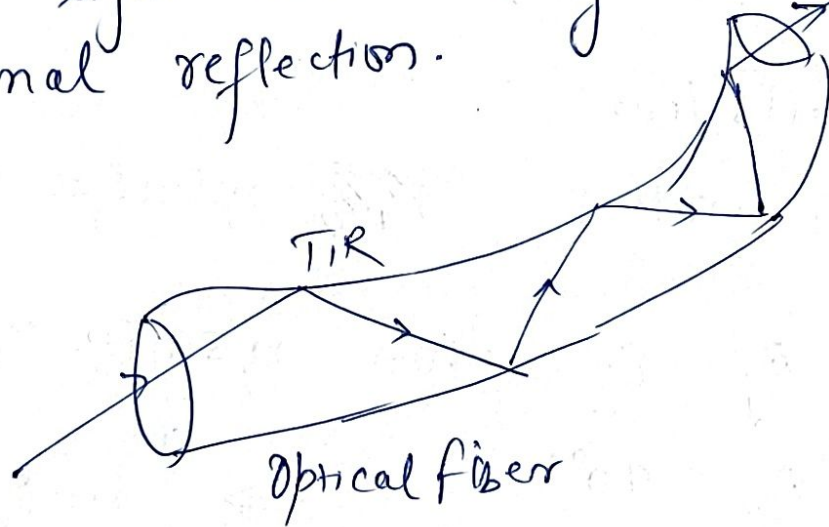


Module 2: Fiber Optics

Fiber optics is a branch of physics that deals with a technology in which signals are converted from electrical signal to optical signal, transmitted through thin glass fiber and reconverted into electrical signal.

* Optical fiber:

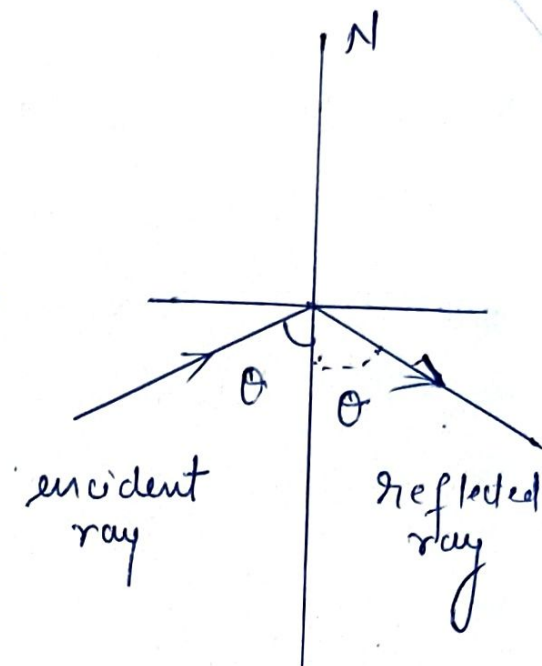
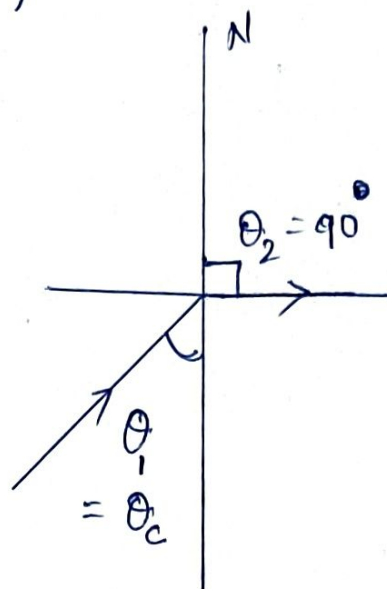
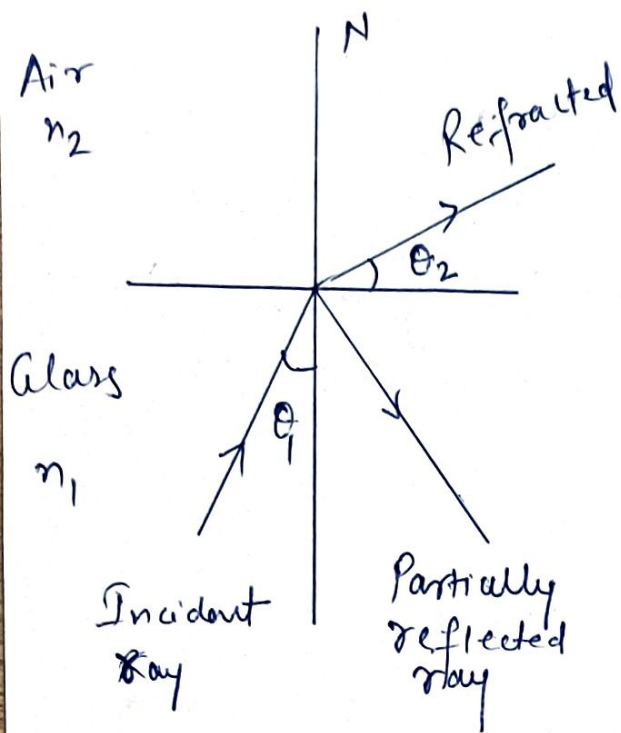
An optical fiber is a cylindrical waveguide made of transparent dielectric (glass/plastic), which guide light waves along its length by total internal reflection.



* Where do we use it

- 1) Internet and Broadband
- 2) Telephone & Communication
- 3) Medical Endoscopy
- 4) Sensors & Aerospace
- 5) Data Centers

* Total Internal Reflection (TIR)



$\theta < \theta_c$: Refracts into rarer medium.

$\theta_1 = \theta_c$: grazes the interface

$\theta > \theta_c$: TIR

Using Snell's law

we can write: ~~$n_1 \sin \theta_1 = n_2 \sin \theta_2$~~ $n_1 \sin \theta_1 = n_2 \sin \theta_2$

when $\theta_1 = \theta_c$ then $n_1 \sin \theta_c = n_2 \sin 90^\circ$

$$\theta_2 = 90^\circ$$

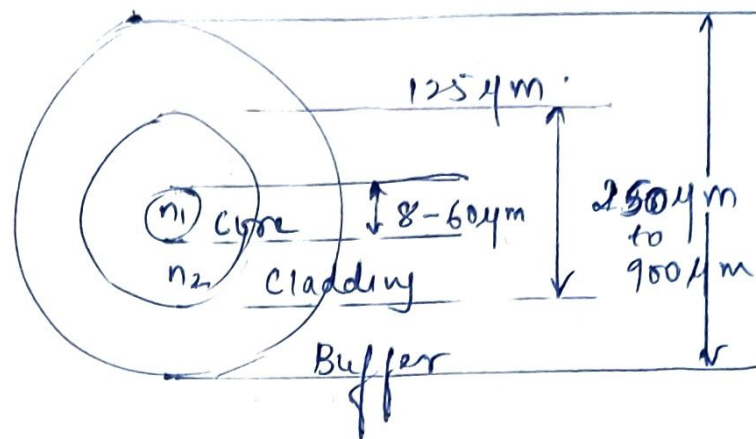
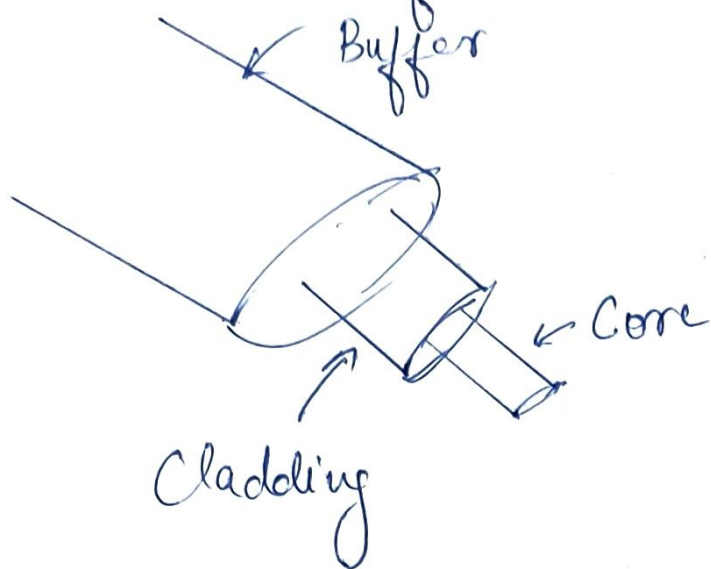
$$\Rightarrow \boxed{\sin \theta_c = \frac{n_2}{n_1}}$$

when the rarer medium is air $n_2 = 1$ & let $n_1 = \mu = n$

$$\therefore \sin \theta_c = \frac{1}{n} \Rightarrow \boxed{\theta_c = \sin^{-1}\left(\frac{1}{n}\right)}$$

Where θ_c is Critical angle i.e. the angle of incidence at which the incident ray undergoes TIR.

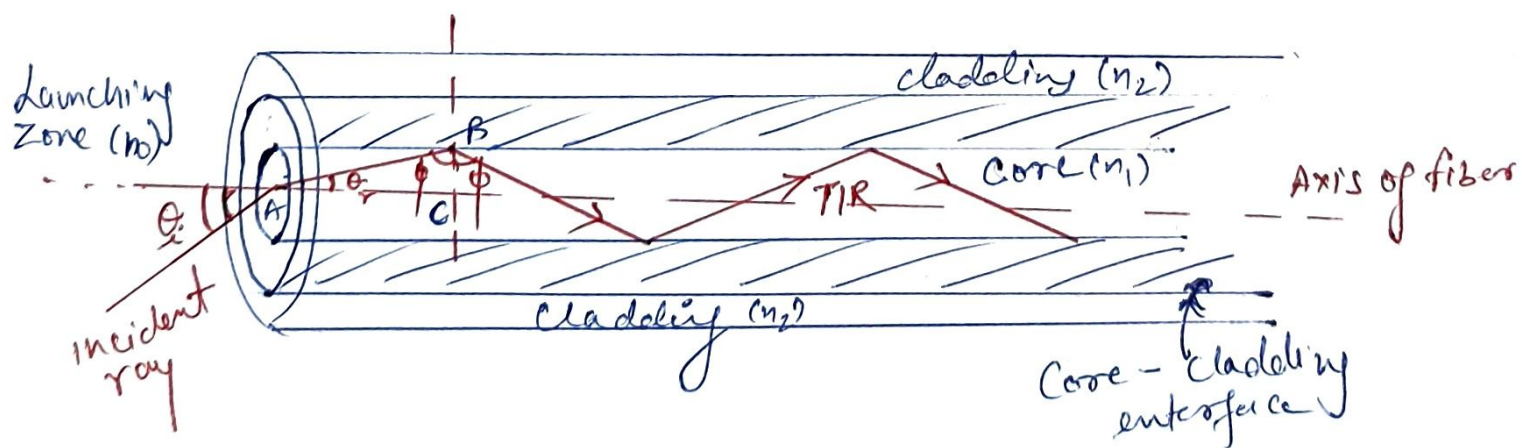
Structure of optical fiber



The optical fiber is Cylindrical in shape and has 3 Co-axial regions

- 1) Core: The innermost Cylindrical region use for guiding light
- 2) Cladding: The middle Coaxial region having refractive index lower than core
- 3) Buffer: The outermost region is called sheath or protective buffer coating

Acceptance angle :



Expression for acceptance angle

- 1) Let the refractive index of core be n_1 and cladding be n_2 .
- 2) Let n_0 be the refractive index from which the light is launched into the core.
- 3) Assume:
 θ_i : angle at which light enters the fiber w.r.t fiber axis
 θ_r : angle of refraction into the core w.r.t axis
 ϕ : the angle at which refracted ray strikes the core-cladding interface
- 4) If ϕ is greater than critical angle (ϕ_c) then the incident ray undergoes total internal reflection as ($n_1 > n_2$). As long as $\phi > \phi_c$ the light will stay within the fiber
- 5) Using Snell's law to the launching face of fiber we get

$$n_0 \sin \theta_i = n_1 \sin \theta_r$$

$$\Rightarrow \sin \theta_i = \frac{n_1}{n_0} \sin \theta_r \quad \text{--- (1)}$$

from $\Delta^{le} ABC$ we can write

$$\begin{aligned} \sin \theta_r &= \sin(90 - \phi) \\ \sin \theta_r &= \cos \phi \quad \text{--- (2)} \end{aligned} \quad \text{as } \left\{ \begin{array}{l} \theta_r = 180 - 90 - \phi \\ \text{sum of all angles} = 180^\circ \end{array} \right.$$

The Critical angle for TIR is given as

$$\sin \phi_c = \frac{n_2}{n_1} \Rightarrow \cos \phi_c = \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$$

$$\therefore \cos \phi_c = \frac{\sqrt{n_1^2 - n_2^2}}{n_1} \quad \text{--- (3)}$$

from (1) and (2) we can write $\sin \theta_i = \frac{n_1}{n_0} \cos \phi$

when $\phi = \phi_c$ then $\sin \theta_i \rightarrow \sin \theta_{i, \max}$

$$\therefore \sin \theta_{i, \max} = \frac{n_1}{n_0} \cos \phi_c \quad \text{--- (4)}$$

from (3) and (4) we can write

$$\sin \theta_{i, \max} = \frac{n_1}{n_0} \left(\frac{\sqrt{n_1^2 - n_2^2}}{n_1} \right) = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

most of the time the incident ray is launched from the air therefore $n_0 = 1$ and $\theta_{i, \max} \approx \theta_0$

Hence

$$\sin \theta_0 = \sqrt{n_1^2 - n_2^2}$$

$$\therefore \theta_0 = \sin^{-1} \left(\sqrt{n_1^2 - n_2^2} \right)$$

θ_0 is called the acceptance angle of the fiber

3

Acceptance angle is the maximum angle that a light ray can have relative to the axis of fiber and propagate down the fiber.

Numerical Aperture:

The numerical aperture (NA) is defined as the sine of the ~~angle~~ acceptance angle.

i.e. $NA = \sin \theta_0 = \sin(\sin^{-1} \sqrt{n_1^2 - n_2^2})$

$$NA = \sqrt{n_1^2 - n_2^2}$$

n_1 : R.I of core
 n_2 : R.I of cladding

* Fractional Refractive Index Change (Δ):

The fractional difference between the refractive indices of the core and cladding is known as fractional refractive index change.

i.e. $\Delta = \frac{n_1 - n_2}{n_1}$ $\left\{ \begin{array}{l} \Delta \text{ always positive} \\ \text{as } n_1 > n_2 \text{ for TIR} \end{array} \right.$

for guiding the light rays effectively $\Delta \ll 1$
($\Delta \approx 0.01$ typically)

Numerical Aperture in terms of Δ

$$N.A = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)}$$
$$= \sqrt{\frac{(n_1 + n_2)}{2} \cdot \frac{(n_1 - n_2)}{n_1} (2n_1)} =$$

We know that $\frac{n_1 - n_2}{n_1} = \Delta$ & approximate $\frac{n_1 + n_2}{2} \approx n_1$

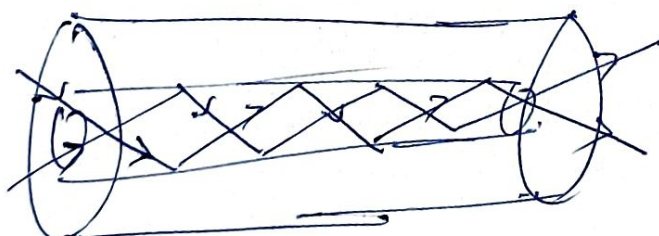
$$\therefore NA = \sqrt{n_1 \Delta (2n_1)} = n_1 \sqrt{2\Delta}$$

$$\therefore \boxed{NA \approx n_1 \sqrt{2\Delta}}$$

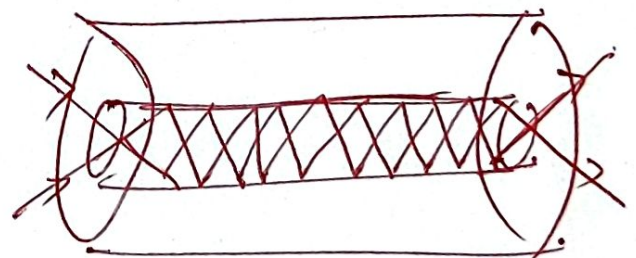
* Physical meaning of N.A.:

1) It is a measure of amount of light that can be accepted by a fiber.

2)



Low N.A.



High N.A.

3) N.A. depends on n_1 and n_2 i.e. R.I of core and cladding and does not depend on the physical dimensions of the fiber

4) A large NA means the fiber will accept -- large amount of light from the source.

5) Generally the values of NA $0.13 < NA < 0.5$